POLITECNICO MILANO 1863 | SCHOOL OF INDUSTRIAL AND INFORMATION ENGINEERING

AEROSERVOELASTICITY OF FIXED AND ROTARY WING AIRCRAFT 2025-2026

Workshop 2: Pitch Flap Aeroelastic Stability Analysis of Puma Helicopter Blade in Hover

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Lecturer:

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1. Introduction and Workflow Description

This report describes an aeroelastic stability analysis of a Puma helicopter blade in hover, focusing on the pitch-flap behaviour of the system. It is based on structural and inertial data provided in Appendix A of [1]. The ultimate goal of this analysis is to construct a stability map of the blade, as a function of pitch link compliance and chordwise position of the blade's center of mass. To that end, different levels of refinement in the analysis, both elastomechanical and aerodynamic, will be employed, comparing their results.

The workflow for all analyses follows the classic template for linear dynamic aeroelasticity analysis. It begins with an integral formulation of the problem, in this case, in terms of the Principle of Virtual Work (PVW). Then, the elastomechanical model is discretised and approximated modal shapes are extracted. These are then used as approximation functions for a Galerkin discretisation of the complete formulation. The dynamic aeroelastic analysis is carried out by assessing the response of the resulting discretised linear system.

The following section will cover the elastomechanical model. Afterwards, different aerodynamic models will be combined with it to assess their differences and construct the desired stability map.

2. Convergence analysis of the elastomechanical model

Throughout this work, the modal analysis of the elastomechanical system will be performed using a Finite Element discretization developed by the author. It is a direct extension of that used in workshop 1 to the new PVW formulation, which includes new terms related to the effect of centrifugal forces acting on the beam and the pitch link stiffness. As such, in the interest of brevity, it will not be detailed here. It suffices to say that a convergence analysis must be performed in order to determine the number of elements required for a good approximation of the modal basis.

This convergence analysis is based on a simple Richardson extrapolation scheme, where the approximation error is predicted by comparing a frequency or mode with its Richardson extrapolation, computed as $\phi(2N_e) - \phi(N_e)$, where $\phi(n)$ represents the quantity of interest, evaluated with a discretisation with n elements. Predicted frequency error is described by error relative to the extrapolated frequency, while the modal error associated to the j-th mode, ϵ_j, is computed with a formula inspired by the L_2 norm:

$$\epsilon_j = \sqrt{\frac{1}{R - x_e}\left(\int_{x_e}^{R} (\hat{w} - \hat{w}_R)^2 \, dx + \int_{x_e}^{R} (\hat{\theta} - \hat{\theta}_R)^2 \, dx\right)}$$

Where x_e is the spanwise station of the flap hinge (assumed coincident with the pitch bearing), R is the radius of the rotor, $\hat{w}$ and $\hat{\theta}$ are the bending and torsion functions associated to the modal shape,

each scaled independently to have unit displacement (bending or torsion) at the tip. The R subscript represents quantities computed with Richardson extrapolation. The scaling is introduced to ensure that the result is non-dimensional.

The evolution of the frequency error for the first six modes is shown in figure 1, while the evolution of modal shape error is shown in 2. In both cases, the error, as expected, decreases as the number of elements increases, even if it doesn't do so monotonically. The number of elements necessary for a given analysis is chosen such that the predicted frequency and modal shape error of all intervening modes is below 0.01 for any number of elements larger than the chosen one. Therefore, in cases where only the first two modes are employed, $N_e = 20$ will suffice, whereas if the first six modes are desired, $N_e = 100$.

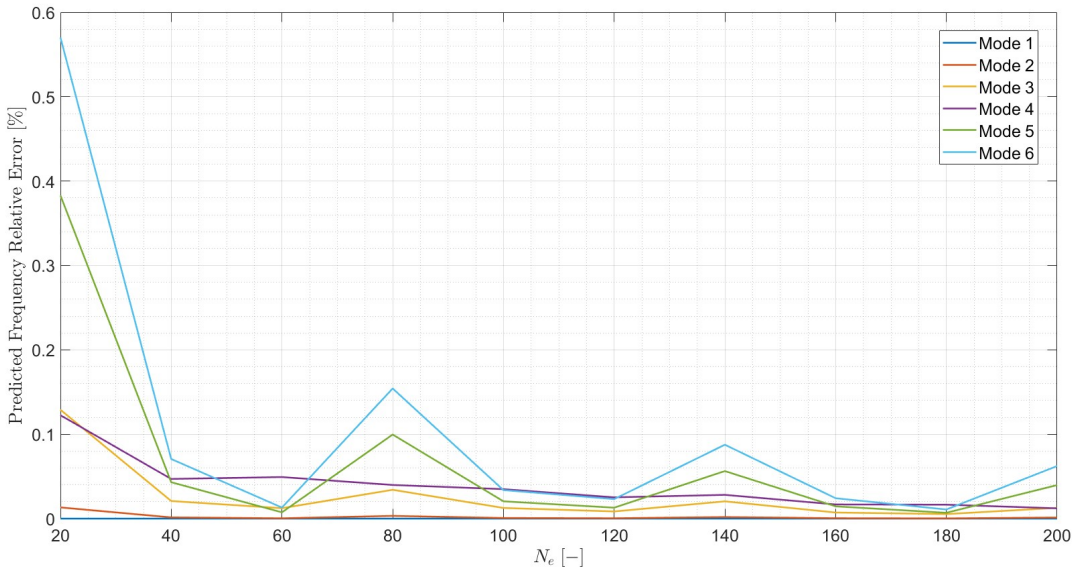

Figure 1: Evolution of predicted frequency relative error

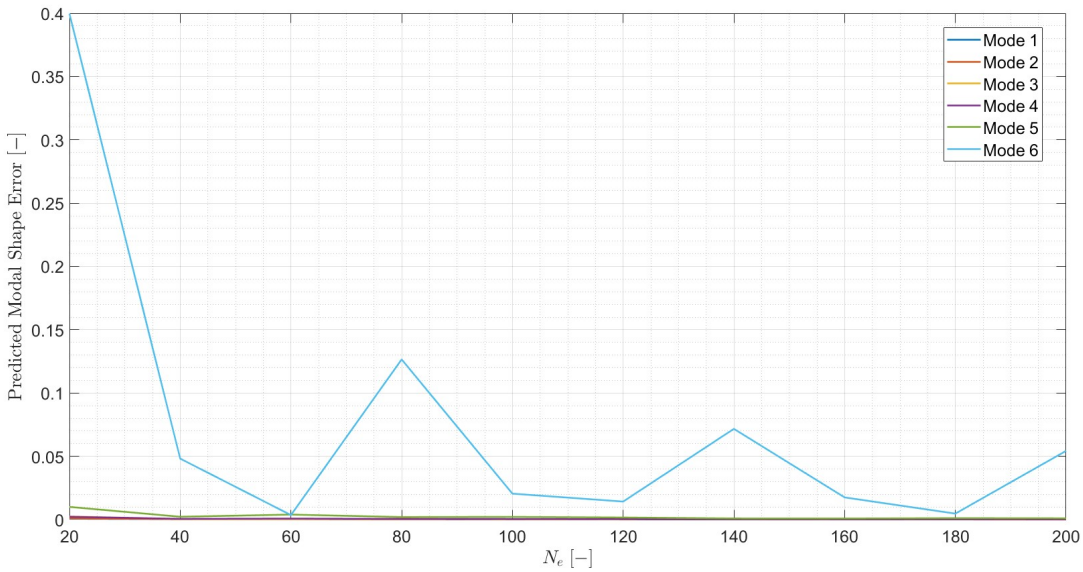

Figure 2: Evolution of predicted modal shape error

3. Exercise 1. Pitch Flap Flutter with Quasisteady Aerodynamics

In this section, the aeroelastic stability of the blade is assessed in nominal operating conditions in hover. The first two modes of the elastomechanical system are taken as approximating functions. These are characterized by frequencies of 29.12 rad/s and 77.67 rad/s, respectively, and proper modal shapes shown in figures 3 and 4.

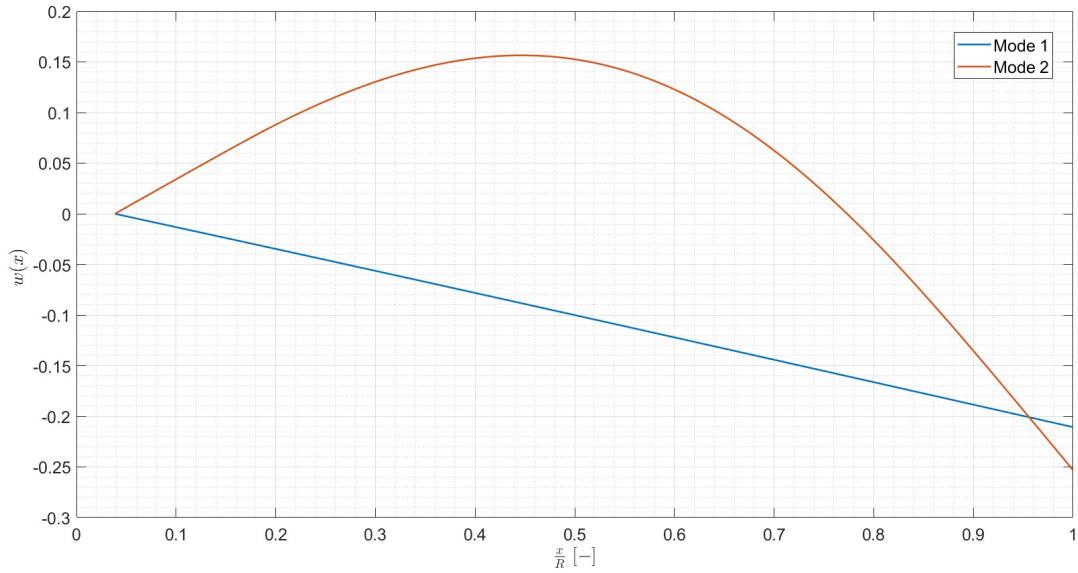

Figure 3: Bending shapes for the first two proper orthogonal modes

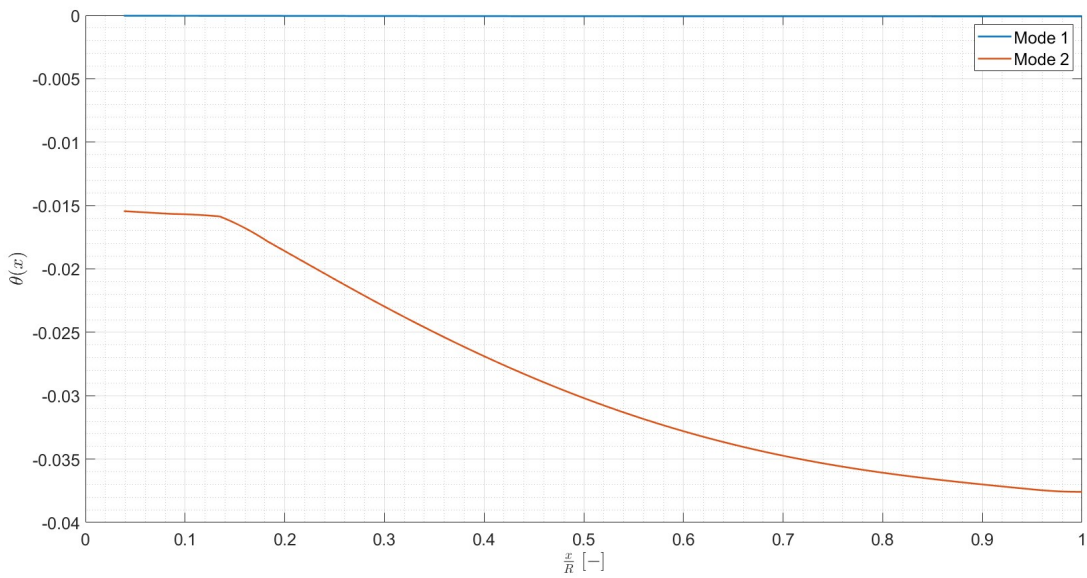

Figure 4: Torsion shapes for the first two proper orthogonal modes

Observing the modal shapes, it is apparent that the first modes represents rigid flapping of the blade, while the second one is a coupled torsion-bending mode. Thus, this combination will be able to capture interplay between flapping and torsion. Note that due to the pitch compliance at x_e, mode shapes can have non-null torsional displacements there.

Aerodynamic loads are modeled with a quasisteady approach. It can be derived from the usual Theodorsen model considering $k << 1$ and taking into account that $C(k \to 0) = 1$. In other words,

this model assumes that the period of the resulting oscillations is much longer than the residence time of the fluid particles. The resulting model is simple to handle and reduces the aeroelastic analysis to a simple eigenvalue problem.

The resulting eigenvalues are $-17.90 + 22.41i$ $[1/s]$ and $-11.92 + 73.70$ $[1/s]$. These correspond to damping ratios of 0.6240 and 0.1597, respectively. Consequently, the system is stable, with a very well damped first mode and a decently damped second mode.

4. Exercise 2: Pitch Flap Flutter using unsteady reference section aerodynamics

In this section, the approximation functions described in the previous section are employed in conjunction with a more refined aerodynamic model. Indeed, the problem is now formulated in terms of Theodorsen aerodynamics and handled via the p-k method. However, the model is still simplified by assuming that k is uniform on the blade and equal to its value at 75% of the blade's span. Note that in practice, the airspeed of a blade section varies linearly with its spanwise location. This approximation is based on the fact that it is the sections around this spanwise station that contribute the most to aerodynamic loading since those inboard of it have low airspeed, while those outboard of it are affected by tip losses.

With the addition of this aerodynamic model, the aeroelastic analysis is no longer a simple eigenvalue analysis, since the aerodynamic matrices will now depend on k, which itself depends on frequency of the structural response. In this project, the most robust solution method has been found to be a continuation algorithm based on the angular frequency of the rotor. Note that the method always uses the modes at nominal frequency as approximating functions. However, it introduces the effect of aerodynamic loads progressively by increasing rotor angular frequency up to the nominal value. In other words, the elastomechanical model is untouched, while the aerodynamic model is made to progressively approach the operating conditions. As a result, the intermediate steps are physically meaningless, but the final result, if converged, is a valid solution of the aeroelastic problem. Throughout the project, 10 steps in Ω, with a maximum of 100 iterations at each step (though in many cases 10 might suffice) have been found more than enough to provide solutions with an absolute error below 10^{-6}.

Applying this algorithm leads to eigenvalues $-16.96 + 25.92i$ $[1/s]$ and $-9.981 + 76.62i$ $[1/s]$. These correspond to damping ratios of 0.5476 and 0.1292, respectively. Therefore, the system is also stable in this model. However, it is important to note that refining the aerodynamic model leads to a 12.24% and a 19.10% reduction in damping ratio, respectively. This reduction implies that the quasisteady approximation is neither accurate nor conservative, which justifies the use of more elaborate aerodynamic models.

5. Exercise 3: Stability Map

The methods shown in the previous section are now extended to construct a stability map of the blade, considering variations in two parameters: the stiffness of the pitch link and the chordwise position of the blade's center of mass. The first is considered ranging from null to its nominal value, while the second is varied between 24% and 40% chord.

The variation of the pitch link stiffness is straightforward, while the variation of the blade's center of mass requires some more discussion. In this case, to isolate the effect of the center of mass and that of the mass distribution, the chordwise position of the section's center of gravity has been displaced, while keeping the section's mass, but considering the effect of the displacement on the section's torsional inertia. Furthermore, this transformation has been deemed more realistic for the aerodynamic portion of the blade, which could be modified more easily than the inboard mechanisms. Therefore, only the center of mass of sections outboard of the aerodynamic cutout has been modified in the creation of the stability map, according to the following formula:

$$Y'_{cg} = Y_{cg} + \delta \, \frac{\int_{x_{co}}^{R} m(x) \, dx}{\int_{x_e}^{R} m(x) \, dx}$$

Where Y_{cg} is the nominal chordwise position of the blade's center of gravity, Y'_{cg} is its desired chordwise position, x_e is the spanwise position of the flap hinge (assumed coincident with the pitch bearing), x_{co} is the spanwise position of the aerodynamic cutout, R is the radius of the blade and δ is a displacement to be uniformly applied to the center of mass of all sections outboard of the cutout, such that $y'_{cg}(x) = y_{cg}(x) + \delta$, where y_{cg} is the nominal chordwise position of the section center of mass and y'_{cg} is its modified chordwise position.

At any rate, this map should be considered as a parametric analysis meant to illustrate the physics of the problem at hand, not as a realistic study of possible modifications to the blade.

Before proceeding with the map, a further modification to the aeroelastic analysis scheme must be discussed. The introduction of centrifugal forces implies that blades with a center of mass sufficiently far from the feathering axis can become statically unstable. The mechanism for this instability is related to the torsion-bending coupling introduced by these forces. In essence, an increase in bending angle leads to an increase in the shear component of the centrifugal force, which leads to an increase in the torsional moment it generates. This, in turn, increases the torsion angle and, with it, the offset between the centrifugal force, applied at the section's center of mass, and the beam axis, increasing the bending moment and, consequently the bending angle. If this loop is strong enough to overcome the beam's inherent bending and torsional stiffnesses, it will render it statically unstable.Annex 1 explores this phenomenon in a simplified model. For the purposes of this map, any such conditions are automatically deemed unsafe. In any case, this phenomenon has always been identified for positions aft of the limit identified by aeroelastic analysis. It might be important, however, if the center of gravity were to be displaced further forward than considered in this project.

Finally, the map itself is constructed following an algorithm that aims to minimize the required aeroelastic analyses. It begins with an equispaced grid of 10 values for the pitch link stiffness and 4 values for the position of the center of gravity. Then, the grid is refined by adding one point on each constant stiffness line, at the middle of the point where the response changes from safe to unsafe. This process is repeated three times to balance accuracy and runtime, but it could be extended to find an arbitrarily small interval on which to place the transition (save for computational errors, evidently).

The resulting stability map is shown in figure 5. Safe combinations are plotted in green, while unsafe ones are colored red. The boundary between the safe and unsafe region is shown in black and is the result of a shape-preserving cubic spline joining the identified points on the frontier. Note that these results are consistent with the theory, where it is seen that, as the control chain stiffness is increased, the safe area grows to include more rearward positions of the center of gravity.

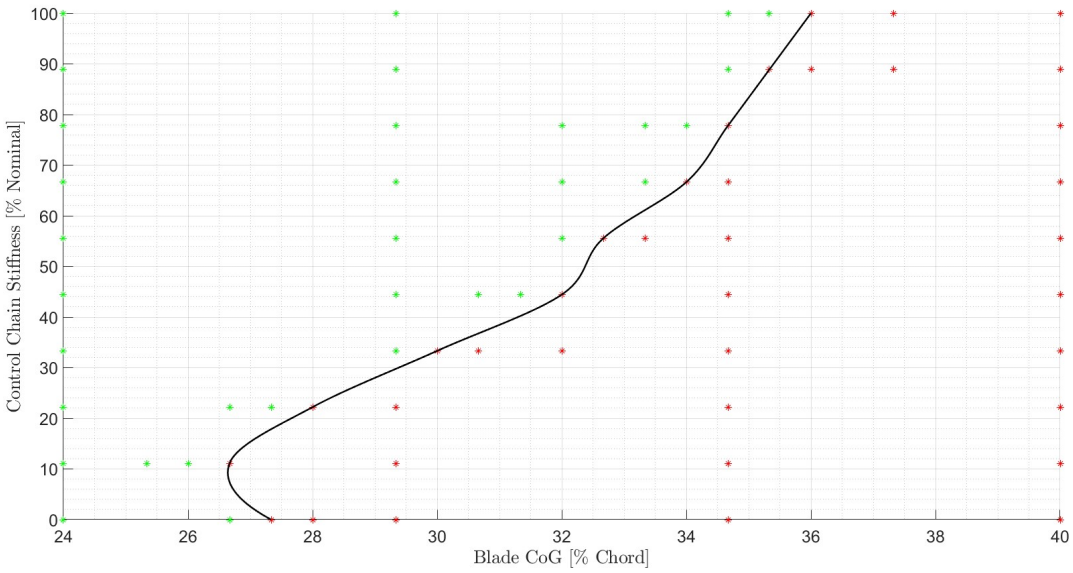

Figure 5: Stability map computed with the first two proper modes and reference section aerodynamics

6. Exercise 4: Stability Map with More Elastic Modes

In this section, the algorithm described in the previous one is extended to account for the first six modes of the elastomechanical system. The resulting stability map is shown in figure 6

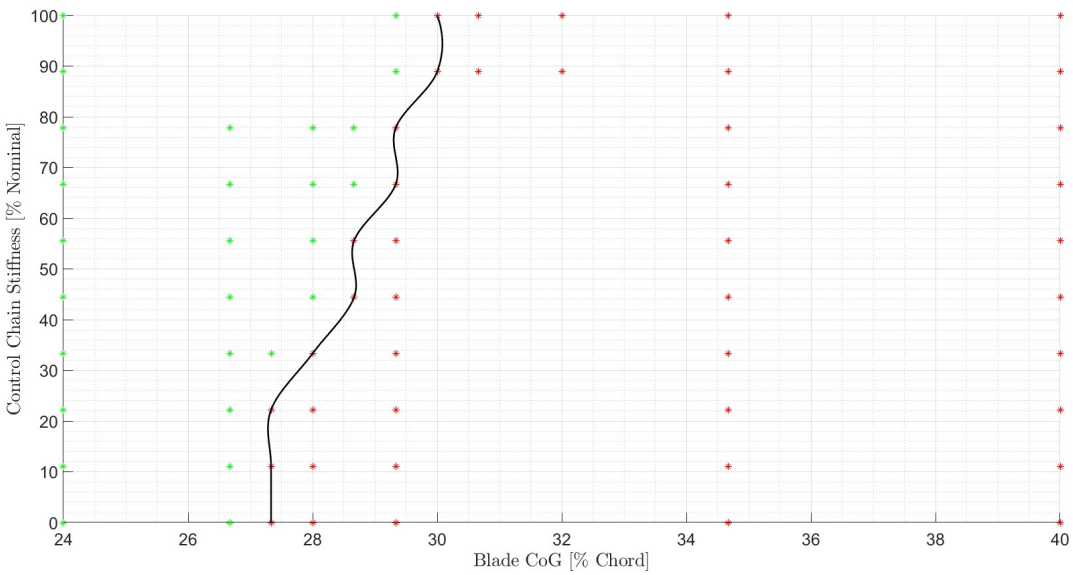

Figure 6: Stability map computed with the first six proper modes and reference section aerodynamics

Observing figure 6, it is immediately apparent that the inclusion of more modes leads to a drastic change in the stability map. Furthermore, as can be seen in figure 7, the prediction with only two modes is not conservative in general. Interestingly, it is conservative for very low control chain stiffness, but quickly becomes very non-conservative as it increases. This shows that instability mechanisms involving more than the first two modes are relevant in this problem and highlights the need for these types of comparison before an analysis is used to take design decisions. In spite of this, at a qualitative level both plots agree in the sense that an increase in the control chain stiffness expands the possible envelope for the chordwise position of the blade's center of gravity backwards.

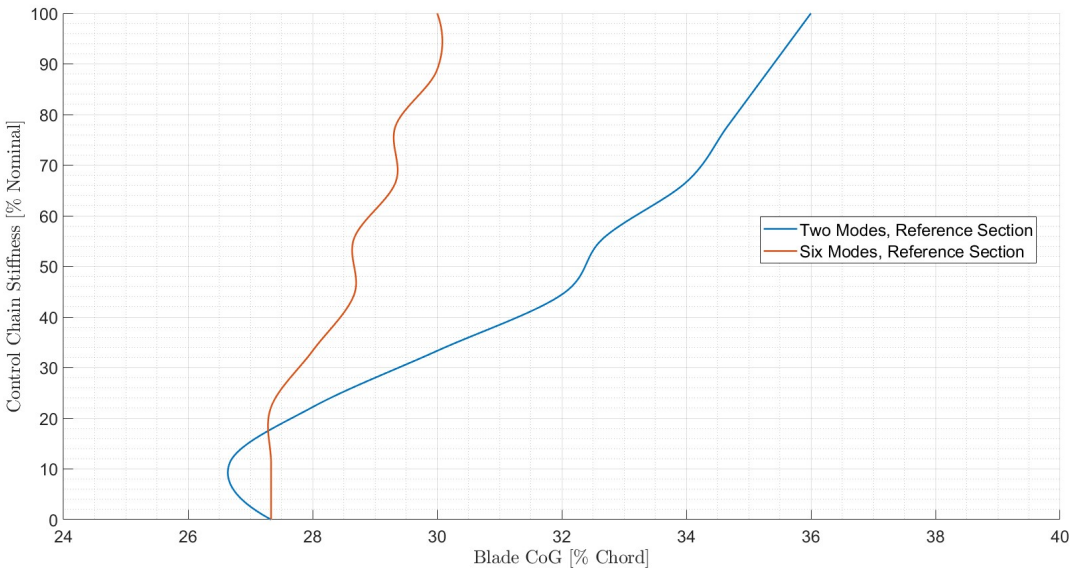

Figure 7: Comparison between stability boundaries for two and six modes in the modal basis, with reference section aerodynamics.

7. Exercise 5: Stability Map with Exact Unsteady Aerodynamics

This section modifies the algorithm in the previous section to account for the variation in reduced frequency along the spanwise stations of the blade. This implies that the aerodynamic matrices are now a function of the spanwise station, which must be accounted for in the modal projection integrals. Beyond this modification, the algorithm is untouched and the resulting stability map is shown in figure 8.

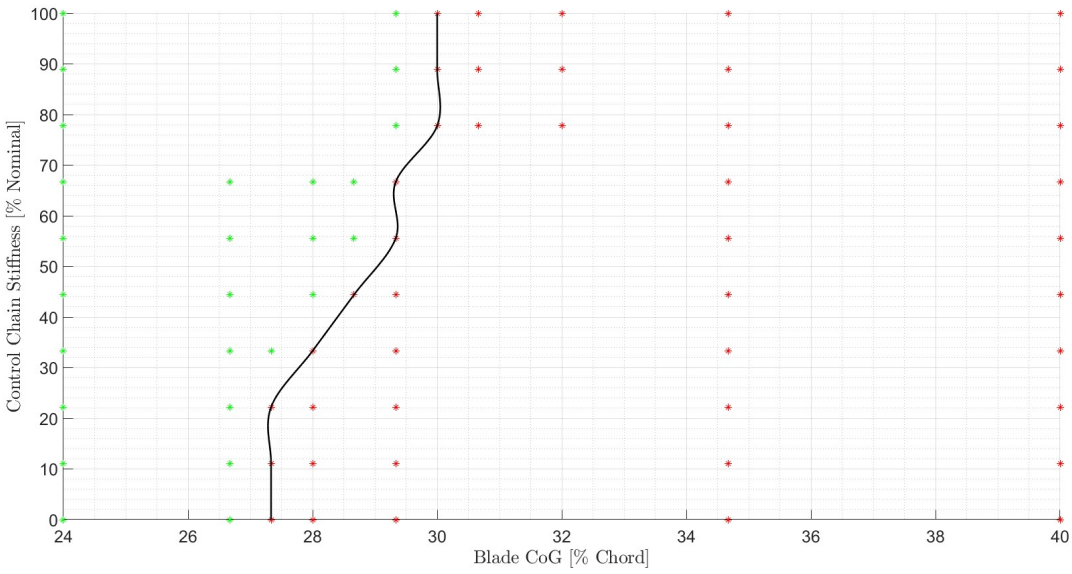

Figure 8: Stability map computed with the first six proper modes and exact unsteady aerodynamics

Observing figure 8, it agrees qualitatively with both of the previously shown stability maps. Quantitatively, it is closer to the one shown in section 6, which implies that the number of modes considered in the analysis has a bigger effect than the change from reference section to exact unsteady aerodynamics. A side-by-side comparison of all maps is shown in figure 9. As expected, the boundaries identified with

six modes are quite similar. For low control chain stiffness, they are identical. For higher stiffness, they generally follow the same trend, except for two bulges where the model with reference section aerodynamics is significantly more conservative. Only on two instances, the model with exact unsteady aerodynamics seems to be slightly more conservative. However, these distances are small enough that they should be deemed comparable, since it is possible that the boundary is slightly distorted by the interpolation and, as such, the situation could be reversed if the grid is refined further.

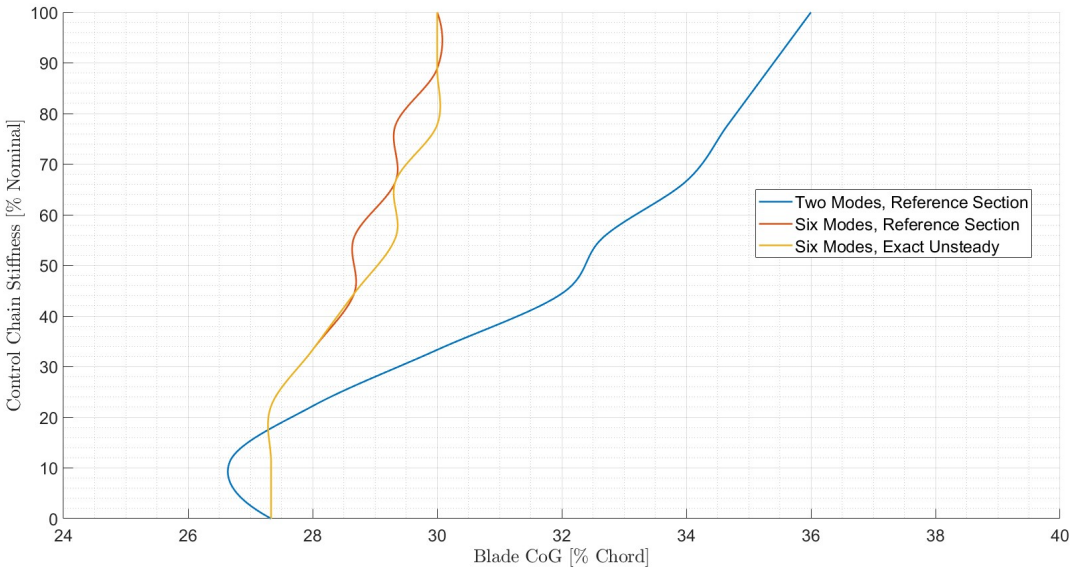

Figure 9: Comparison between stability boundaries for all cases considered.

In conclusion, the number of modes considered in the analysis is shown to determine the shape of the stability boundary more than the aerodynamic model. Consequently, a convergence study based on this parameter is paramount before any results can be deemed trustworthy. Regarding the aerodynamic model, reference section aerodynamics is generally conservative and provides some noticeable simplification of the analysis, reducing computational cost. Therefore, it can be an interesting choice depending on the design phase. On the other hand, the exact unsteady model allows a designer to push the envelope further if desired, which can justify its use. Nevertheless, it should be borne in mind that "pushing the envelope" in practice will be more demanding on all components of the analysis, not just the aerodynamic model, since a reduction of the safety margin must be accompanied with a reduction in the prediction's uncertainty to maintain adequate levels of risk.

8. Annex 1: Static Instability in a Simplified Model

This annex expands on the static instability commented in section 5. To highlight the mechanism for instability, consider a simplified model, consisting of a rigid flat plate connected to the axis of rotation via a single point with an elastic link of stiffnesses k_φ and k_θ in bending and torsion, respectively. The plate has a mass distribution such that I_{xx}, I_{yy} and I_{xy} are the second moments of inertia about the connection point. Note that the following convention has been chosen for I_{xy}:

$$I_{xy} = -\int_{Plate} \sigma xy \, dxdy$$

Where σ represents the mass per unit surface of the plate. The resulting stiffness matrix for a linearized system with the bending and torsion angles φ and θ, respectively, as generalized coordinates is:

$$K = \begin{bmatrix} K_{\varphi\varphi} & K_{\varphi\theta} \\ K_{\theta\varphi} & K_{\theta\theta} \end{bmatrix} = \begin{bmatrix} k_\varphi + \Omega^2 I_{yy} & \Omega^2 I_{xy} \\ \Omega^2 I_{xy} & k_\theta + \Omega^2 I_{xx} \end{bmatrix}$$

The determinant of this matrix is given by:

$$det(K) = \left(k_\varphi + \Omega^2 I_{yy}\right)\left(k_\theta + \Omega^2 I_{xx}\right) - \Omega^4 {I_{xy}}^2$$

Given that the trace of the matrix is positive, the matrix will not be positive definite when its determinant is null or negative, which leads to the following condition:

$$\left(k_\varphi + \Omega^2 I_{yy}\right)\left(k_\theta + \Omega^2 I_{xx}\right) - \Omega^4 {I_{xy}}^2 \leq 0$$

Note that if $I_{xy} = 0$, the condition is impossible to meet, since the left hand side is the product of two positive numbers.

This condition can be expanded to form a quadratic polynomial in Ω^2:

$$\left(I_{xx}I_{yy} - I_{xy}^2\right)\Omega^4 + \left(I_{yy}k_\theta + I_{xx}k_\varphi\right)\Omega^2 + k_\varphi k_\theta \leq 0$$

The roots of this polynomial are real if, and only if:

$$\left(I_{yy}k_\theta + I_{xx}k_\varphi\right)^2 - 4k_\varphi k_\theta\left(I_{xx}I_{yy} - I_{xy}^2\right) \geq 0$$

This condition can be rewritten as:

$$\left(I_{yy}k_\theta - I_{xx}k_\varphi\right)^2 + 4k_\varphi k_\theta I_{xy}^2 \geq 0$$

Which is always true for $k_\varphi \geq 0$, $k_\theta \geq 0$. The roots of the equation can be written as:

$$\Omega^2 = \frac{-\left(I_{yy}k_\theta + I_{xx}k_\varphi\right) \pm \sqrt{\left(I_{yy}k_\theta + I_{xx}k_\varphi\right)^2 - 4k_\varphi k_\theta\left(I_{xx}I_{yy} - I_{xy}^2\right)}}{2\left(I_{xx}I_{yy} - I_{xy}^2\right)}$$

Note that only positive values for Ω^2 make physical sense. Based on this result, the behaviour of the system can now be classified depending on the sign of the leading term in the polynomial.

If $\left(I_{xx}I_{yy} - I_{xy}^2\right) < 0$, then the system will be statically unstable for Ω^2 outside of the interval limited by the roots of the aforementioned polynomial. Thus, the system will be unstable for:

$$|\Omega| \geq \sqrt{\frac{-\left(I_{yy}k_\theta + I_{xx}k_\varphi\right) - \sqrt{\left(I_{yy}k_\theta + I_{xx}k_\varphi\right)^2 - 4k_\varphi k_\theta\left(I_{xx}I_{yy} - I_{xy}^2\right)}}{2\left(I_{xx}I_{yy} - I_{xy}^2\right)}}$$

Note that the absolute value on Ω is consistent with the fact that static stability will not be affected by the system rotating counter-clockwise or clockwise.

On the other hand, if $\left(I_{xx}I_{yy} - I_{xy}^2\right) > 0$, it is easy to see that both solutions for Ω^2 are negative. Since, in this case, the determinant is negative for Ω^2 between both roots, this implies that the system is stable for all positive, i.e., physically meaningful, Ω^2.

Furthermore, if $\left(I_{xx}I_{yy} - I_{xy}^2\right) = 0$, the leading term of the polynomial vanishes and it is apparent that the inequality cannot be verified for positive Ω^2.

In conclusion, this simplified model illustrates the fact that, under adequate conditions on the mass distribution, the effect of centrifugal forces, particularly their coupling of bending and torsion, can lead to a statically unstable system for angular velocities over a certain threshold, dependent on the inertial and stiffness characteristics of the system.

References

[1] William G. Bousman et al. A comparison of lifting-line and cfd methods with flight test data from a research puma helicopter. Technical report, NASA, 1996.